

2MTEST : A semi-automatic tool for legacy test migration to MTEST

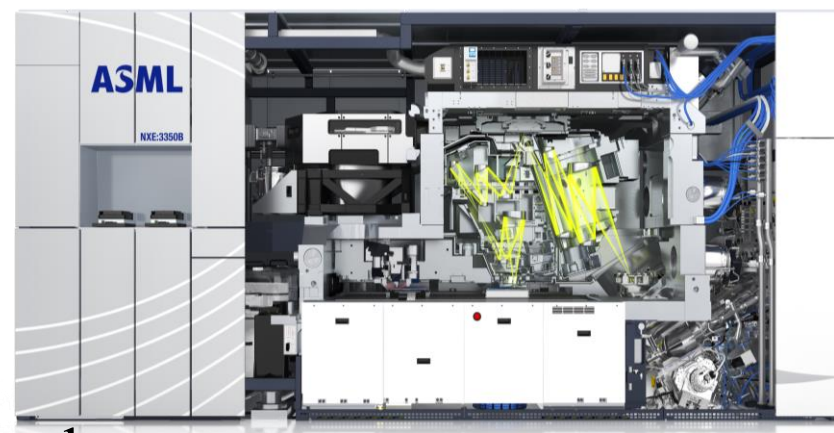
Introduction/ Context

Legacy Test Code:

- Initializing the whole twin scan software.
- Slow test run and high CPU usage

MTEST:

- ASML specific Module Testing framework.
- Facilitate testing in controlled and isolated environment
- Only necessary components are activated



Problem Statement



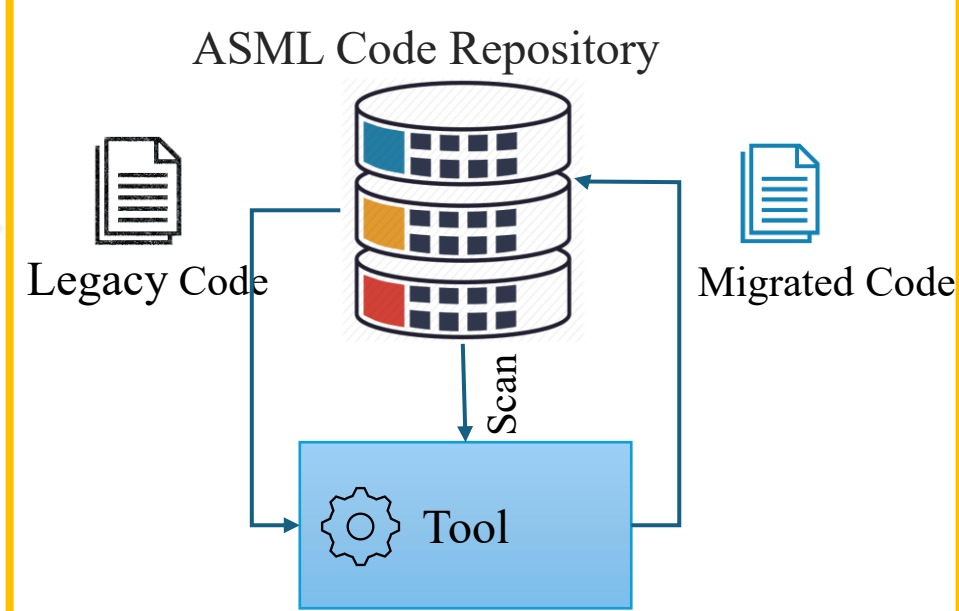
Legacy Test Codes
C, C++ & Python

Manual Migration

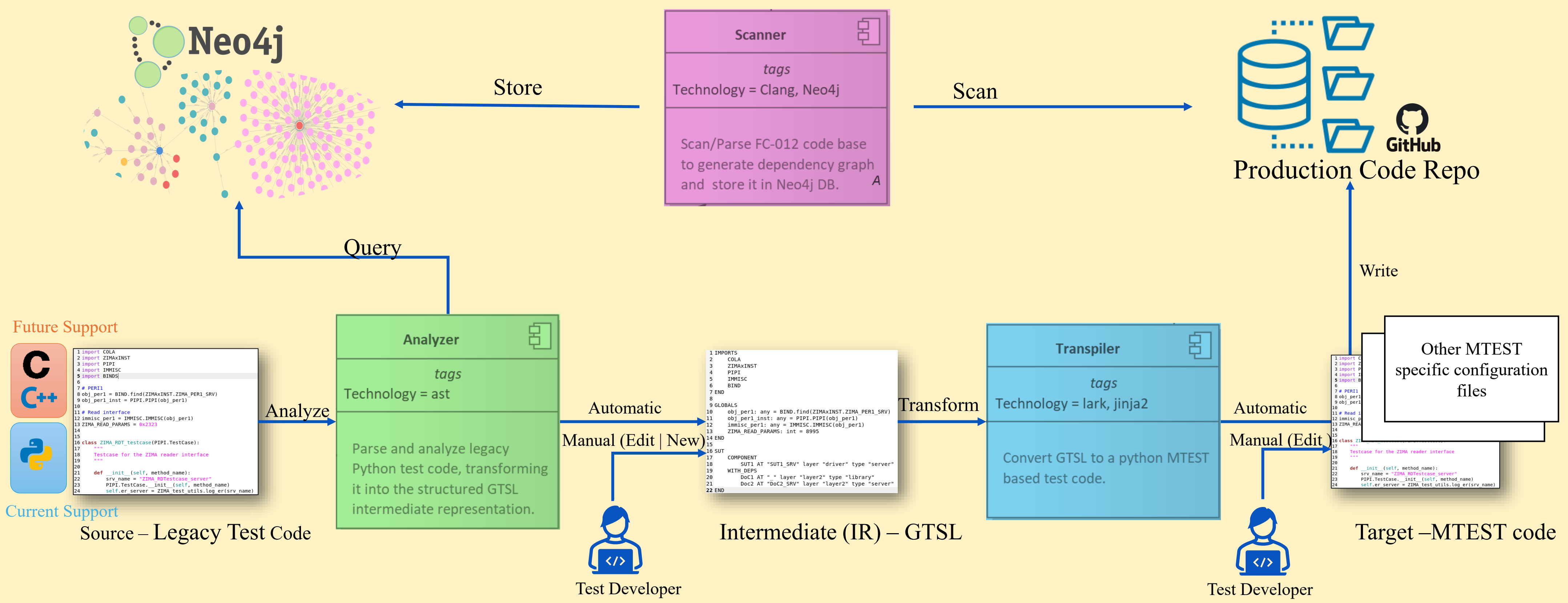
MTEST

- Time-consuming
- Error-prone
- Costly

Solution Automation



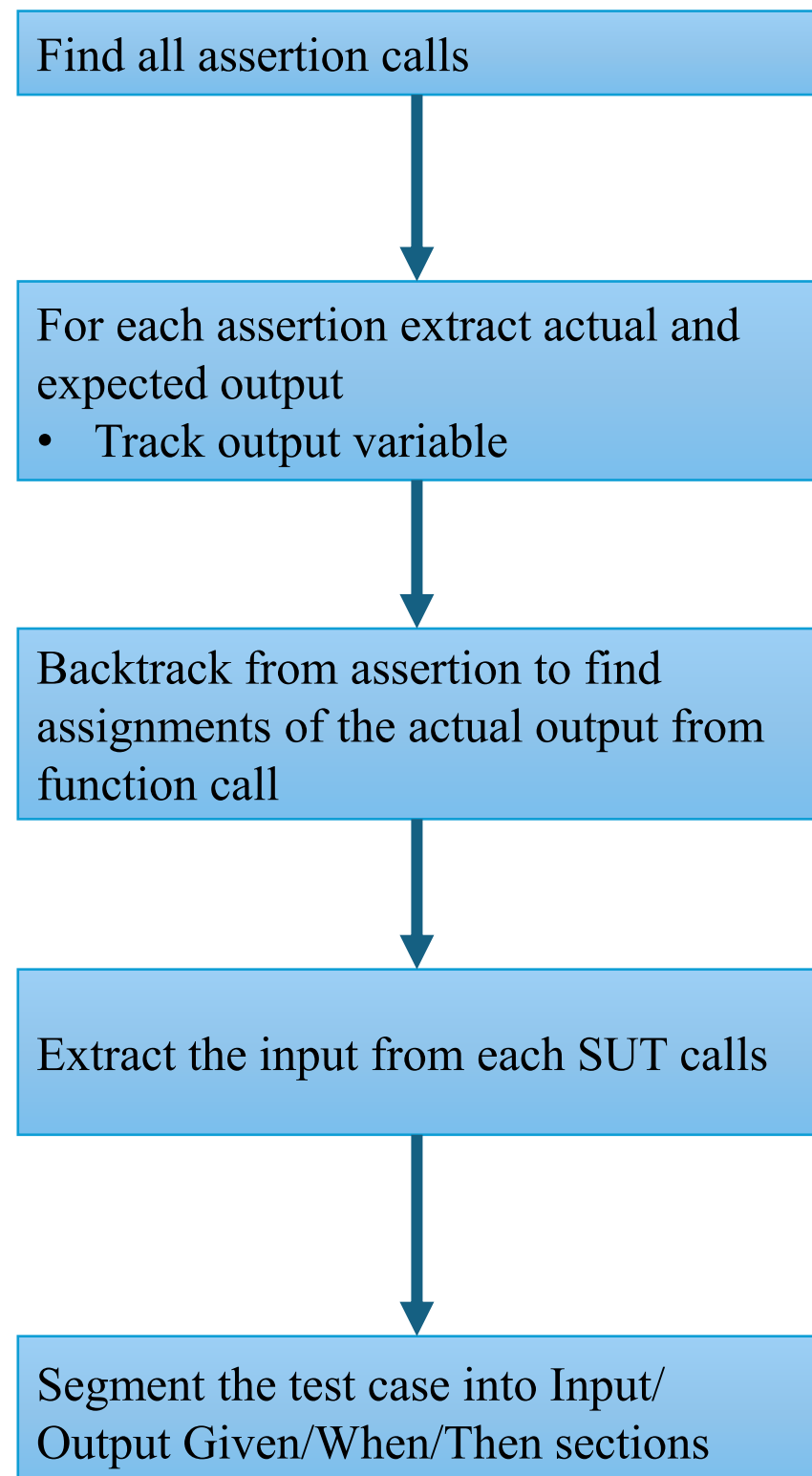
Modular Migration Pipeline



Scanner: Knowledge Graph

- Pre-Processing:** Sanitizes `compile_commands.json`.
- Hybrid Parsing:** Parsing C/C++ source code and ASML specific various configuration files using `libclang` and `regex`.
- Node/Hierarchical Discovery:** Creates a node for every entity (**Component**, **File**, **Function**, **Interface**) and build a global function definitions **symbol table**.
- Dependency Linking:** Creates relationships (**:CALLS**, **:INCLUDES**, **:PART_OF**) using symbol table.
- Output:** knowledge graph stored in Neo4j DB.
 - Cypher** pseudo query => "Given *SUT* function, what external components and interface are ultimately called?"

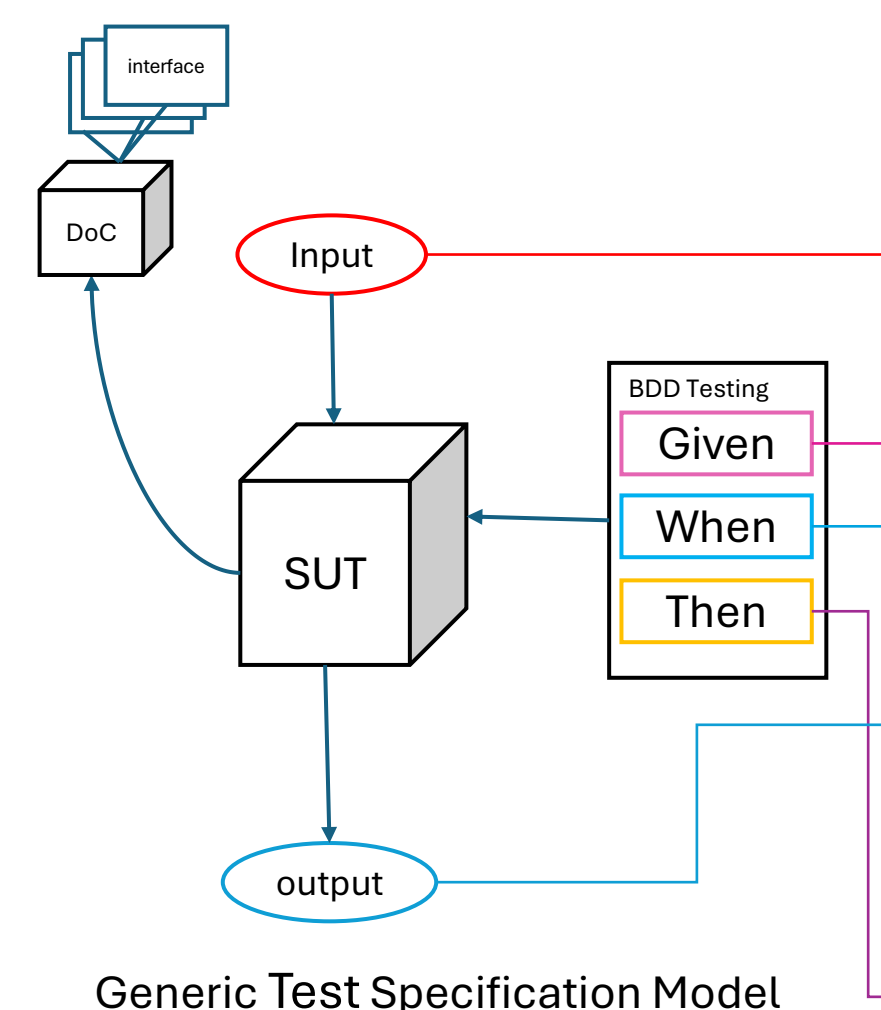
Analyzer: Algorithm



Transpiler: GTSL – DSL

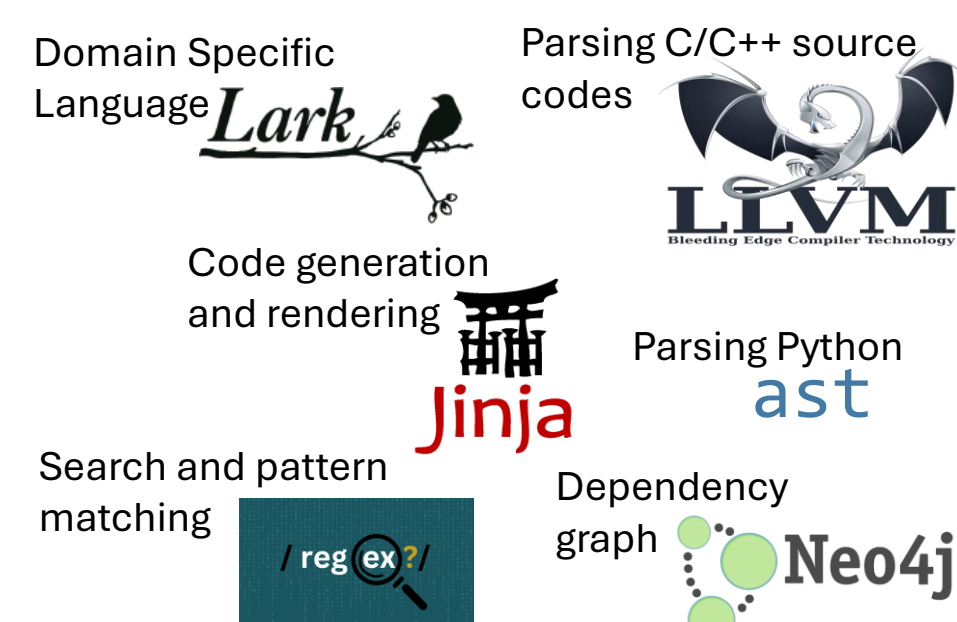
Generic Test Specification Language (GTSL)

- A domain-specific language (DSL) designed as an **intermediate representation** between legacy test code and the target MTEST based test.



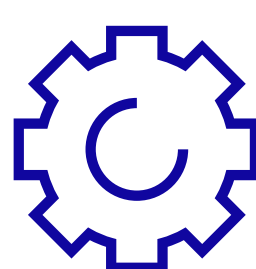
```
IMPORTS
  CXXIF
  IPPE
END
GLOBALS
  global_int: int = 123
END
SUT
  COMPONENT
    sut1 at "server_addr" layer "layer1" type "server"
  WITH DEPS
    doc1 at "server_addr" layer "layer2" type "server"
    doc2 at "." layer "layer3" type "library"
  END
END
DOUBLE
  INTERFACE MOKA FROM CNOT AT "SRV_ADDR1" IFTYPE "library"
  INTERFACE STUB1 FROM ZIMAXINST AT "STUB_ADDR" IFTYPE "mock"
END
SUITE TestSuiteExample
  SETUP
    call obj_perl_inst.terminate()
  END
  TEARDOWN
    set global_counter = 0
    comment "Clean up test environment"
  END
  HELPER
    FUNCTION check_status(obj: any) RETURNS bool
      return call obj.status()
    END
  CASE test case sum
    INPUT
      in reader elem: any
      in scramble: int[] = [12,3,4]
    END
    OUTPUT
      out expected_out_1: float
    END
    GIVEN
      set temp = input1
      call obj_perl_inst.initialize(PIPI.INIT_LEVEL_1)
    END
    WHEN
      set result = call compute_sum(input1, input2)
    END
    THEN
      EXPECT_EQUAL result, 30
    END
  END
END
```

Technologies Used



Value Added /Result

Prototype Tool

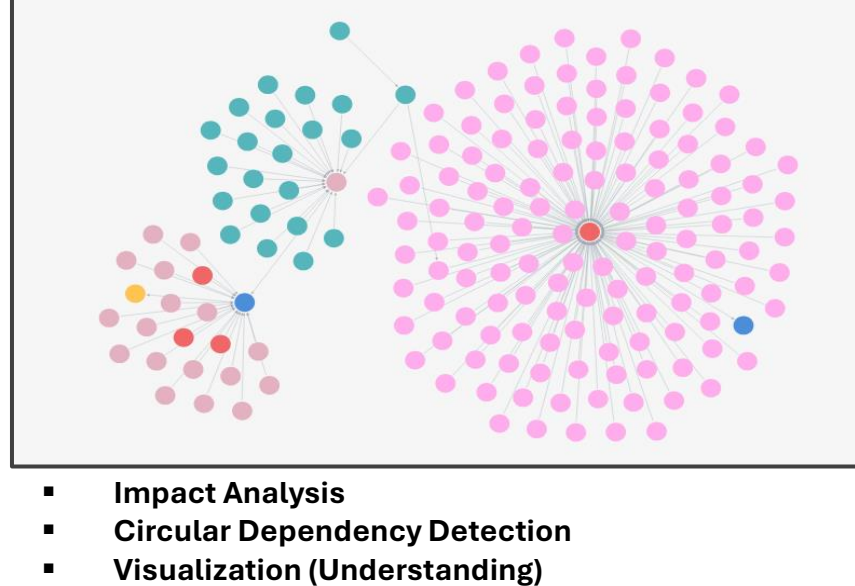


A tool that cuts 2 – 5 hours of manual migration into less 20 seconds

Domain Specific Language

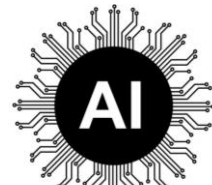
```
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97  ZIMAXINST
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99  ZIMAXINST
100 ZIMAXINST
```

Dependency Graph



- Impact Analysis
- Circular Dependency Detection
- Visualization (Understanding)

Future Direction



Replacing the heuristic-based analyzer with AI/ML model.



Extending GTSL as test framework

